

Personal, Relevant Background, and Future Goals Statement

Applicant: James Wade

Personal Statement and Motivation: When I knelt beside a woman in a wheelchair in a small Ecuadorian clinic, adjusting the engineered pressure mat my team had developed, I saw engineering as more than equations and prototypes. The clinician explained to me that uneven pressure distribution can lead to painful sores and infections for wheelchair users who cannot feel their lower body. As she shifted and the sensor map responded in real time, her caregiver began to understand how to prevent any harm before it started. At that moment, I realized that engineering could directly and personally enhance quality of life. It grounded me in a belief that has shaped every step of my engineering path: **technology matters most when it meaningfully improves people's lived experiences**. I learned that the most elegant solution means little if it doesn't reach the people who need it — a lesson that drives (1) my focus on robust, deployable systems rather than purely theoretical advances, and (2) my goal of becoming a university professor.

I gained this experience through a year-long humanitarian product design program at Brigham Young University in May 2024. **I led a team that developed a cost-effective pressure mat for a non-profit wheelchair clinic in Ecuador**, leveraging earlier research from another lab [1] and applying it in a real-world setting. Commercial pressure mats can cost over \$10,000; our open-source alternative, built with conductive fabric and an Arduino-based sensor array, costs under \$60. The clinic now uses it to evaluate custom wheelchair cushions and reduce pressure injuries. This project taught me that **impactful engineering requires not only technical innovation but also a deep understanding of deployment constraints — whether it be cost, reliability, or communication limitations**. This insight directly informs my proposed research on communication-adaptive safety for multi-robot systems in resource-constrained environments.

My commitment to improving lives began before Ecuador. After my freshman year at BYU, I spent a year teaching English in Mongolia (2021-2022), where I saw firsthand how education expands opportunity. I paused my studies to volunteer, and the impact of providing knowledge to others has stayed with me. Teaching students with limited resources showed me how mentorship can unlock potential that circumstances might otherwise constrain. Just as I helped Mongolian students access new opportunities through English, **I aim to become a professor who helps talented students from all backgrounds achieve their potential through research mentorship**. This commitment drives both my current mentoring of 20+ undergraduate researchers and my vision for leading a lab that develops engineers based on merit and dedication.

Relevant Background and Formative Experiences: My research journey began in BYU's microfluidics lab and has evolved into parallel tracks in robotics and control: I lead research in the CMR Lab on inflatable robots for lunar exploration while also pursuing safety-critical control research in the MAGICC Lab. These experiences, plus internships at two national labs and leadership roles across campus, have prepared me for graduate research in safe multi-robot coordination and for my goal of becoming a professor.

Early Research Foundation: Beginning in BYU's microfluidics lab, **I co-authored a publication in the International Journal of Heat and Mass Transfer as a second-year undergraduate** [2]. This early exposure to rigorous peer review taught me that impactful research requires not only novel ideas but also clear communication, a skill I now apply when writing grant proposals and mentoring students. I carried that passion with me into the Compliant Mechanisms and Robotics (CMR) Lab, where I worked with Mark Rober to develop The World's Smallest Nerf Gun, contributing to the design and prototyping of the mechanism. **This project has reached over 75 million views on YouTube** and continues to help inspire future engineers. This large-scale outreach demonstrated how creative engineering communication can inspire the next generation—a lesson that now drives my outreach efforts and my commitment to making robotics accessible.

Current Robotics Research — CMR Lab (June 2023 - present): My longest and most formative research experience has been in the Compliant Mechanisms and Robotics (CMR) Lab, where I led the development of a modular inflatable robot designed for lunar exploration (Figure 1). I joined the lab in June 2023 as one of three initial undergraduate researchers working with Dr. Nathan Usevitch. Over the next year, **I helped grow the group as team lead to more than twenty active researchers**. This growth was not coincidental — as project manager, I developed onboarding protocols, created modular sub-projects that allowed new researchers to contribute immediately, and established a culture where undergraduates could lead significant

technical efforts at a time when there were no graduate students yet in the lab. **Through this experience, I learned that successful research requires not just individual skill but the ability to build and coordinate teams where each member’s strengths amplify the whole.**

The technical challenges of this project have directly prepared me for my proposed research on multi-robot coordination and for pursuing a PhD. Each active roller unit in Figure 1 acts as an independent agent influencing the overall motion of the robot, making the robot a natural multi-agent system. To coordinate these agents, I developed a hierarchical radio communication architecture and iteratively redesigned the roller units to improve actuation reliability. During testing, we confronted a critical question: *what happens if a roller fails?* This prompted me to design a fault-tolerant control protocol that preserves as much of the robot’s reachable workspace as possible in real time — a practical lesson in robustness under uncertainty. Through these challenges, I gained expertise in graph theory, optimization, robust control, and constraint maintenance, and came to appreciate that elegant theory must hold even when a single link—or radio—drops out. This realization directly motivates my proposed research on Communication-Adaptive CBFs that maintain safety guarantees despite network unreliability.

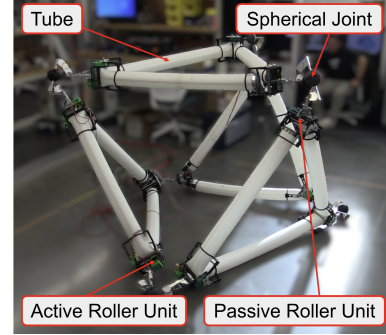


Figure 1: Inflatable truss robot.

Our work’s impact extends beyond technical contributions. **We were selected as one of six national finalists in the NASA BIG Idea Challenge, and I presented our work at three major conferences — LSIC (Lunar Surface Innovation Consortium), the BIG Idea Challenge conference, and UCUR (Utah Conference for Undergraduate Research). I co-authored a paper, currently in process of submission, to *IEEE Transactions on Mechatronics* [3] and will submit for review a first-author paper on workspace maximization under actuator failures by December 2025. I also led grant writing efforts that secured \$140,000 in NASA funding,** learning to articulate not only technical specifications but societal impact. Additionally, I personally received a \$2,000 Utah NASA Space Grant Consortium fellowship in May 2025 for my workspace maximization research. Writing these proposals taught me that successful research requires compelling storytelling that connects technical innovation to human needs.

Understanding that research impact extends beyond publications, **our team organized visits to local elementary schools with 3D-printed robot models.** During BYU’s ‘Lab-o-ween,’ **we used our robot to hand out candy to visiting trick-or-treaters.** Watching a seven-year-old’s eyes widen as our robot moved showed me that accessible demonstrations can spark lasting interest in STEM. As a future professor, I plan to continue this outreach, knowing that today’s amazed child could become tomorrow’s robotics researcher.

Current Safety-Critical Control Research — MAGICC Lab (August 2024 - present): Building on my experience with distributed coordination and fault-tolerant control in the CMR Lab, I sought to deepen my understanding of safety-critical control by joining the MAGICC Lab (BYU’s premier drone laboratory) to study Control Barrier Functions (CBFs) for real-time navigation. In this work, I implement and analyze optimization-based CBF controllers on ground robots and drones, **demonstrating my ability to conduct rigorous, graduate-level research on safety-critical control in real-time environments.** I have shown initial results achieving provable safety using LiDAR-based obstacle detection within a real-time framework, and I plan to submit a first-author journal paper by February 2026. Maintaining real-time feasibility under computation and communication limits is essential for multi-agent systems in disaster-response environments. As a junior in high school, I participated in Hurricane Harvey cleanup in Houston, TX, and I saw the importance of timely and reliable disaster response for those affected — directly motivating my proposed GRFP research on Communication-Adaptive CBFs. This work not only aligns closely with my graduate research goals but also strengthens the theoretical foundation I will build upon in graduate school.

Engineering Internships — Sandia and LLNL: My internships at Sandia and Lawrence Livermore National Laboratories strengthened my ability to implement complex autonomous systems. At Sandia (Summer 2025), I developed a ROS 2 multi-agent autonomy stack for drones and ground vehicles. At LLNL (Summer 2024), I built an STM32-based RC controller handling full hardware-software integration. These experiences **deepened my expertise in system integration, hardware development, and real-time troubleshooting—**

essential skills for multi-agent autonomy research.

Academic and Leadership Excellence: Alongside my research, I have demonstrated academic rigor in my coursework and leadership among my peers. **I have maintained a 4.0 major and overall GPA, including courses in controls, dynamics, deep learning, and optimization.** As the **teaching assistant for a graduate-level robotics course of 60 students**, I lead review sessions, support labs, and help students bridge theory and implementation. As the **elected team lead for the 34-member BYU Mars Rover team**, I coordinate engineering efforts across members from electrical, mechanical, and computer engineering, as well as computer science and geology. I deliver progress reports and conduct ROS training for new members. These experiences have strengthened my ability to communicate complex ideas, manage collaborative projects, and mentor peers — skills that will directly enhance my impact as a graduate researcher and educator.

Intellectual Merit:

In total, my preparation for graduate study integrates theoretical understanding, research leadership, and practical implementation across robotics and control. I have contributed to research in soft robotics, multi-agent autonomy, and safety-critical control, with experience ranging from system modeling and optimization to hardware deployment. My work in the MAGICC Lab on Control Barrier Functions directly supports my proposed research, and my internships at Sandia and Lawrence Livermore provided experience translating algorithms into operational systems. **These experiences provide a strong foundation for contributing new knowledge in communication-adaptive safety mechanisms for multi-robot systems, as outlined in my research statement, and in becoming a well-rounded researcher.**

Broader Impacts:

Dissemination of Knowledge and Educational Infrastructure: **I have communicated my research contributions through four co-authored papers** that have been published or will soon be submitted for review, as well as presentations in competitive national research settings. These experiences demonstrate my ability to effectively share scientific knowledge with both academic and professional audiences. In graduate school, I plan to develop open-source implementations of Communication-Adaptive CBFs and create educational resources to help students and other users engage with safe autonomy methods (details in research statement).

Societal Well-Being: I am committed to ensuring that the benefits of robotics research extend to those who need them most. My work in Ecuador demonstrated how cost-effective and thoughtful engineering solutions can improve the quality of life. Similarly, my year teaching in Mongolia reinforced the importance of education. As a future professor, I aim to advance technology that supports critical operations in disaster response and planetary exploration while fostering research communities that value mentorship and collaboration.

STEM Education: **I have actively mentored undergraduate researchers, led collaborative lab environments, visited local elementary schools, and worked as a teaching assistant for a graduate robotics course.** These experiences reflect my ongoing commitment to increasing access to engineering education and inspiring the next generation of students to participate in STEM. Through these efforts, I aim to increase participation in robotics and empower students to contribute meaningfully to technological innovation.

Future Goals: I plan to pursue a PhD in robotics and controls to advance safety-critical coordination for multi-robot systems under uncertain communication. My immediate goal is to extend Control Barrier Functions to adapt to dynamic communication conditions, enabling provably safe deployment in real-time multi-robot systems, capable of saving lives in time-critical scenarios. I aim to work with faculty who bridge theoretical guarantees with hardware validation, such as Zac Manchester at MIT and his work on optimal control. Long-term, I intend to become a professor advancing multi-robot autonomy in high-stakes environments like disaster response and space exploration. I will lead a research group that develops robust autonomy methods while training students to translate safe control theory into practical, real-world systems. Ultimately, as a graduate student, researcher, and aspiring professor, I aim to create life-improving autonomous technologies — ones that make meaningful differences — and inspire the next generation of engineers.

References [1] Nathan Day et al., *Journal of Micromechanics and Microengineering*, 2018; [2] Derek Sanchez et al., *International Journal of Heat and Mass Transfer*, 2025; [3] Mihai Stanciu et al., *In Process of Submission to IEEE Transactions on Mechatronics*, 2025;